

Ch. 17 - Thermochemistry Studyguide

KEY CONCEPTS

17.1 The Flow of Energy—Heat and Work

- Heat always flows from a warmer object to a cooler object.
- A system gains heat in an **endothermic process** and loses heat in an **exothermic process**.
- Heat flow is measured with two common units: the **calorie** and the **joule**.
- The heat capacity of an object depends on both its **mass** and its **chemical composition**.

17.2 Measuring and Expressing Enthalpy Changes

- In calorimetry, the heat released by a system equals the heat absorbed by its surroundings. Conversely, the heat absorbed by a system equals the heat released by its surroundings.
- The enthalpy change for a reaction can be treated like any other **reactant or product** in a chemical equation.

17.3 Heat in Changes of State

- The heat absorbed by a melting solid is exactly the same as the heat lost when the liquid solidifies; that is, $\Delta H_{\text{FUSION}} = -\Delta H_{\text{SOLIDIFICATION}}$
- The heat absorbed by a vaporizing liquid is exactly the same as the heat lost when the vapor condenses; that is, $\Delta H_{\text{VAPORIZATION}} = -\Delta H_{\text{CONDENSATION}}$
- Heat is either released or absorbed during the formation of a solution.

17.4 Calculating Heats of Reaction

- You can calculate the heat of a reaction by applying **Hess's law of heat summation** or by using **standard heats of formation**.

VOCABULARY:

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|---------------------------------|--------------------------------|
| ● Calorimeter | ● Molar Heat Of Condensation |
| ● Chemical Potential Energy | ● Molar Heat Of Fusion |
| ● Endothermic Process | ● Molar Heat Of Solidification |
| ● Enthalpy | ● Molar Heat Of Solution |
| ● Exothermic Process | ● Molar Heat Of Vaporization |
| ● Heat | ● Specific Heat |
| ● Heat Capacity | ● Standard Heat Of Formation |
| ● Heat Of Combustion | ● Surroundings |
| ● Heat Of Reaction | ● System |
| ● Hess's Law Of Heat Summation | ● Thermochemical Equation |
| ● Law Of Conservation Of Energy | ● Thermochemistry |

KEY EQUATIONS:

Specific Heat:

$$C = \frac{q}{m \times \Delta T}$$

Heat of system: $q_{sys} = \Delta H = -q_{surr} = -m \times C \times \Delta T$

Standard heat of reaction: $\Delta H^0 = \Delta H_f^0(\text{products}) - \Delta H_f^0(\text{reactants})$

Review Questions

1. How does the heat of vaporization for a substance compare to its heat of condensation?
2. What is the difference between heat capacity and specific heat?
3. If a reaction container feels cold to the touch, is the process endothermic or exothermic? Why?
4. According to Hess's law, how can you determine the enthalpy change of a reaction that is too difficult to measure directly?
5. A **50.0g** piece of copper ($C = 0.385 \text{ J/g}^\circ\text{C}$) is heated to **100°C** and then placed into a calorimeter. If the copper cools down to **25°C**, how much heat (in Joules) did the copper release?

- Please also review other self-guided practice sheets, notes, and homework -